

Web Design Basics

Lecture 04

Topics:

- Design Considerations
 - Resolution and Load time
 - Measuring web performance Metrics
 - Core Web Vitals - Google's Performance Metrics
 - Web Metrics for web performance & Tracking
 - Website Organization
 - Web Pages (Layout) Design



Design Considerations

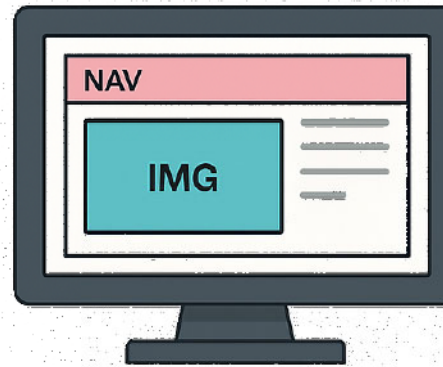
- **Resolution and Load time**
- **Web Site Organization & Web Pages Design**

Screen Resolution

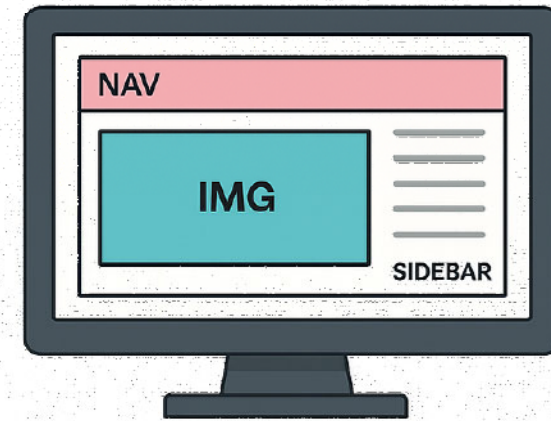
- Test your design at various screen **resolutions** (800×600, 1024×768, 1280×1024, and modern sizes like 1920×1080)
- Keep page content **centered** for consistent layout



800x600
Compact layout

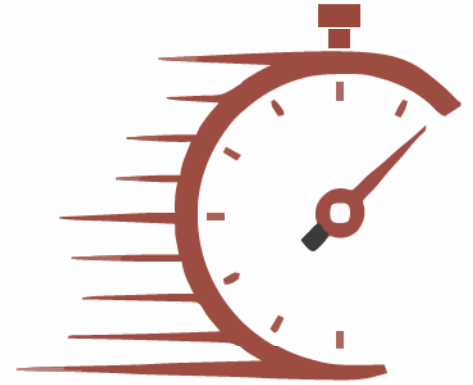


1024x768
Standard desktop layout



1280x1024
Wide screen layout

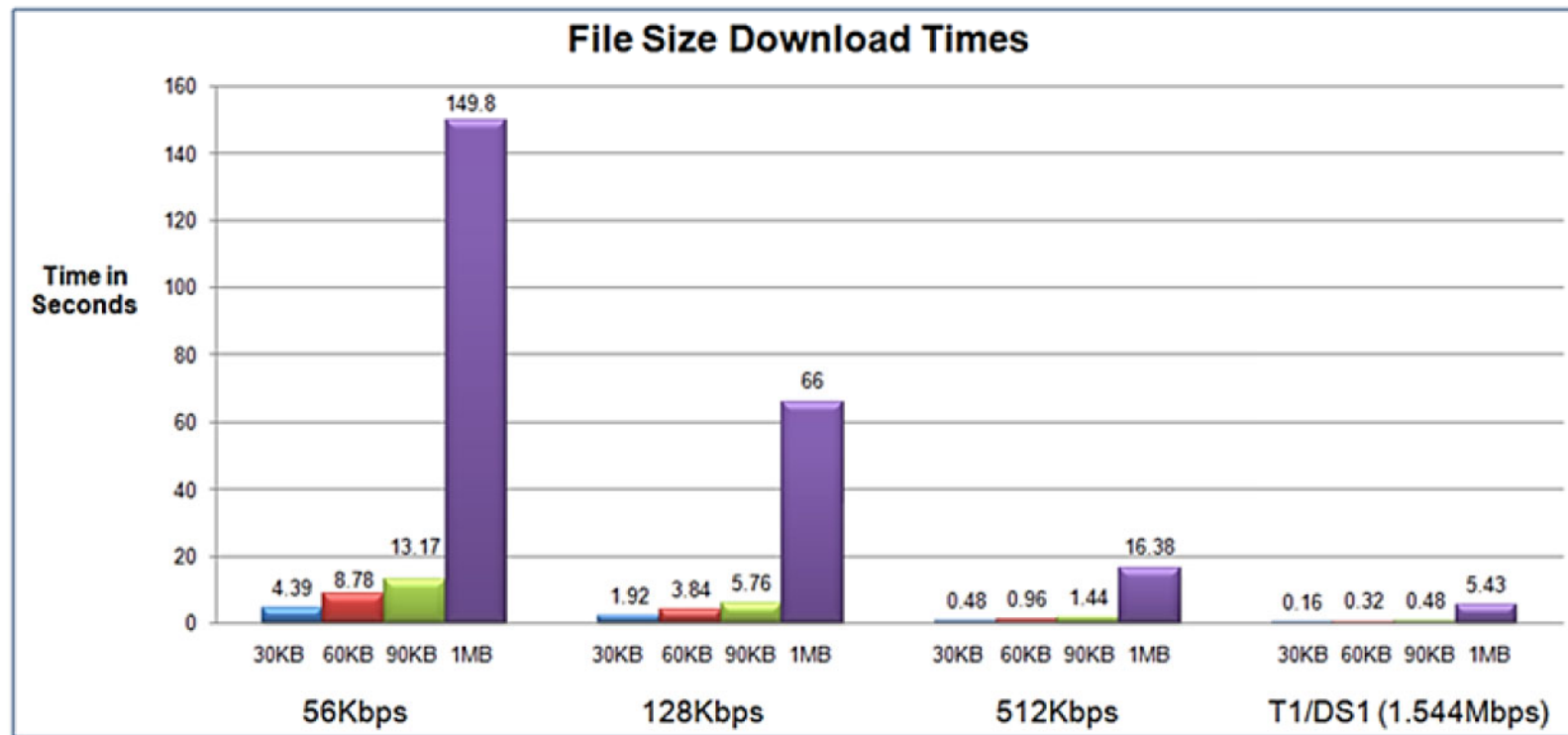
- Use adaptive widths:
 - **Fixed width**: consistent layout (good for desktops)
 - **Percentage width**: responsive layout (adapts to window size)
- Preview on multiple devices
 - desktop, tablet, mobile



Load time considerations

Web Page Design Load Time

- Measure the **load time** of your pages (benchmark is available)
- Try to limit web page document and associated media to under **60K** on the home page



 **AVERAGE PAGE LOAD TIME**
IN SECONDS

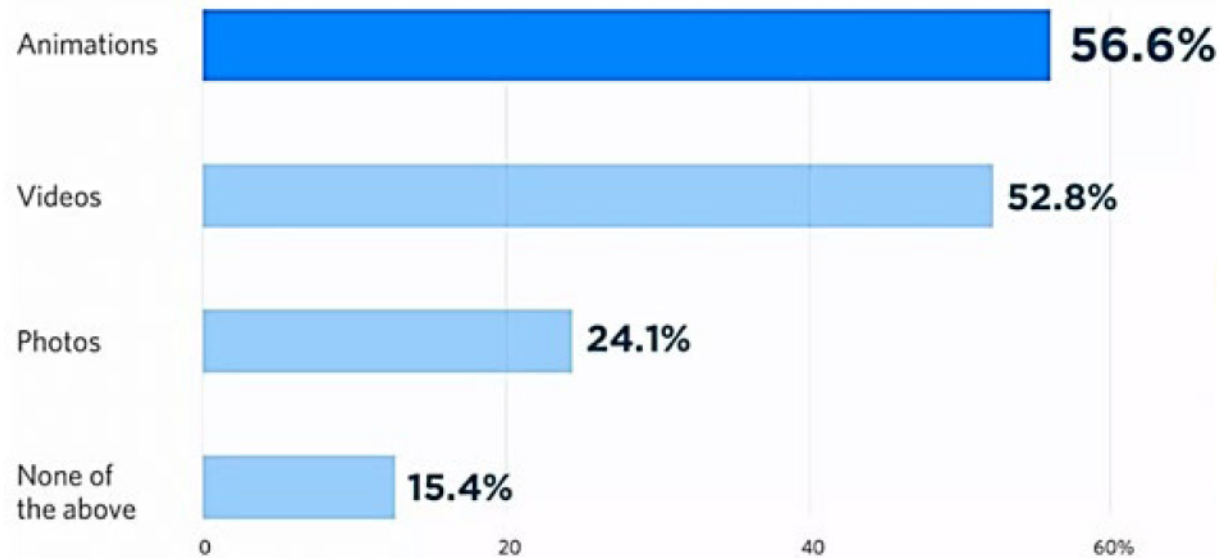


The average page
load time for a
webpage is **3.21**
seconds

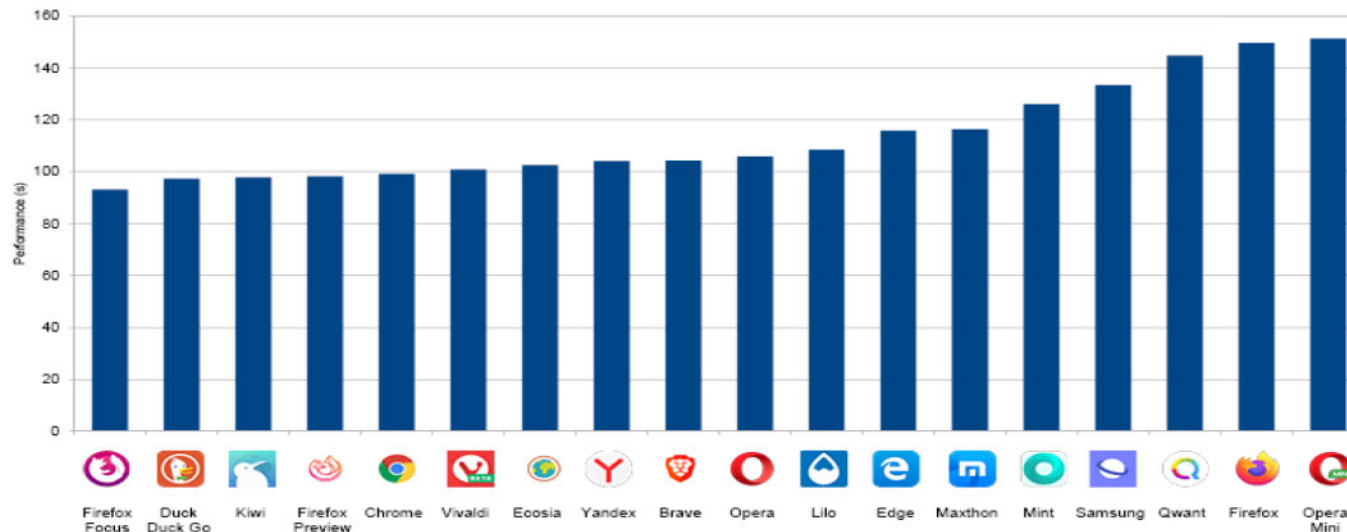
Factors Affect the Web Page Load Time

Imagery effect

Performance profile in Chrome



Browser effect



Technologies effect

Based on the analysis of 150 000 websites:

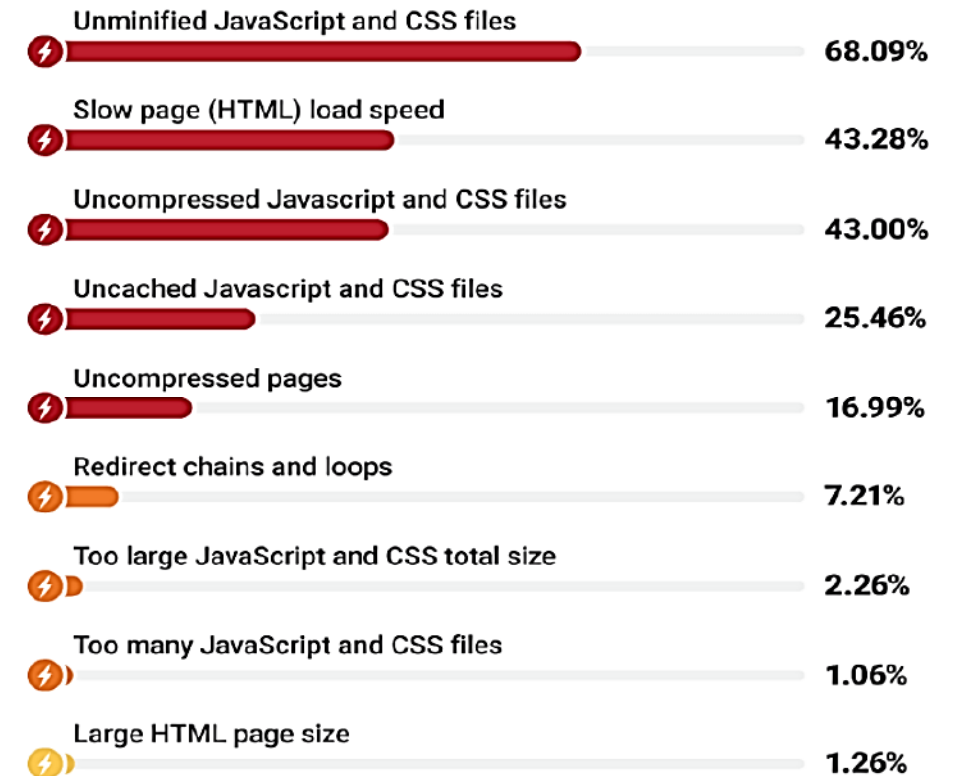
82.89% HAVE ISSUES THAT NEGATIVELY AFFECT PAGE SPEED

All issues listed below have a negative impact on website performance, but their severity levels vary.

SEVERITY LEVELS



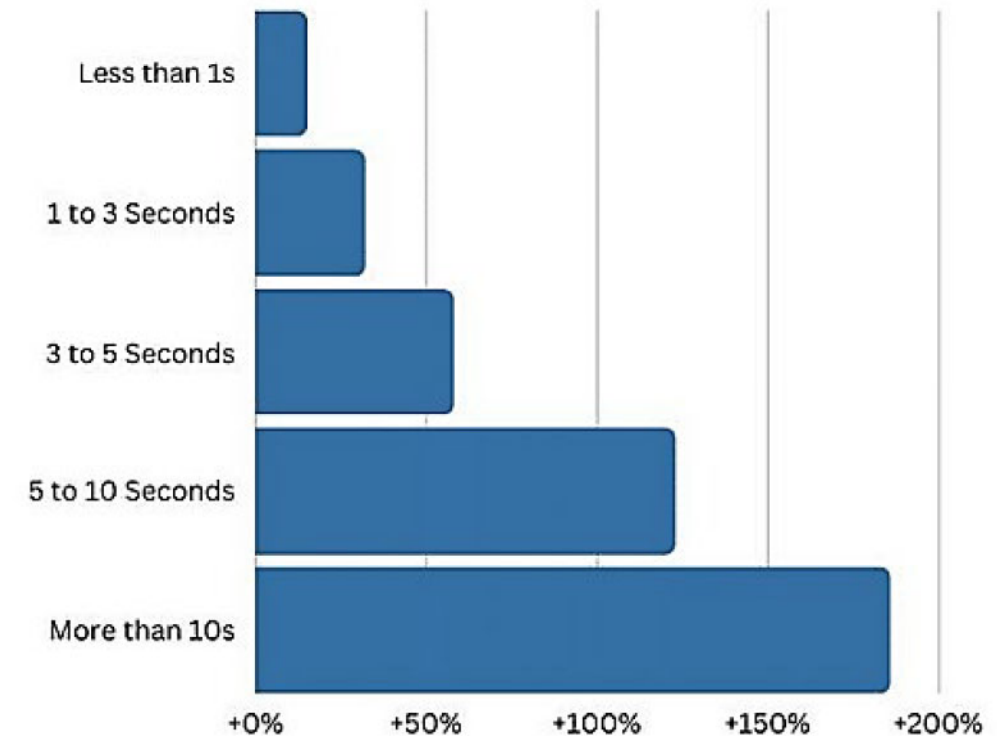
Below you can see the percentage of websites with specific issues:



Measuring web performance Metrics

Why Web Performance Matters

- 53% of mobile users **abandon** sites that **take >3 seconds to load**
- Google uses **page speed** as a ranking factor (SEO impact)
- Amazon found every **100ms delay costs 1% in sales**



Fast websites = Better user experience + Higher conversions

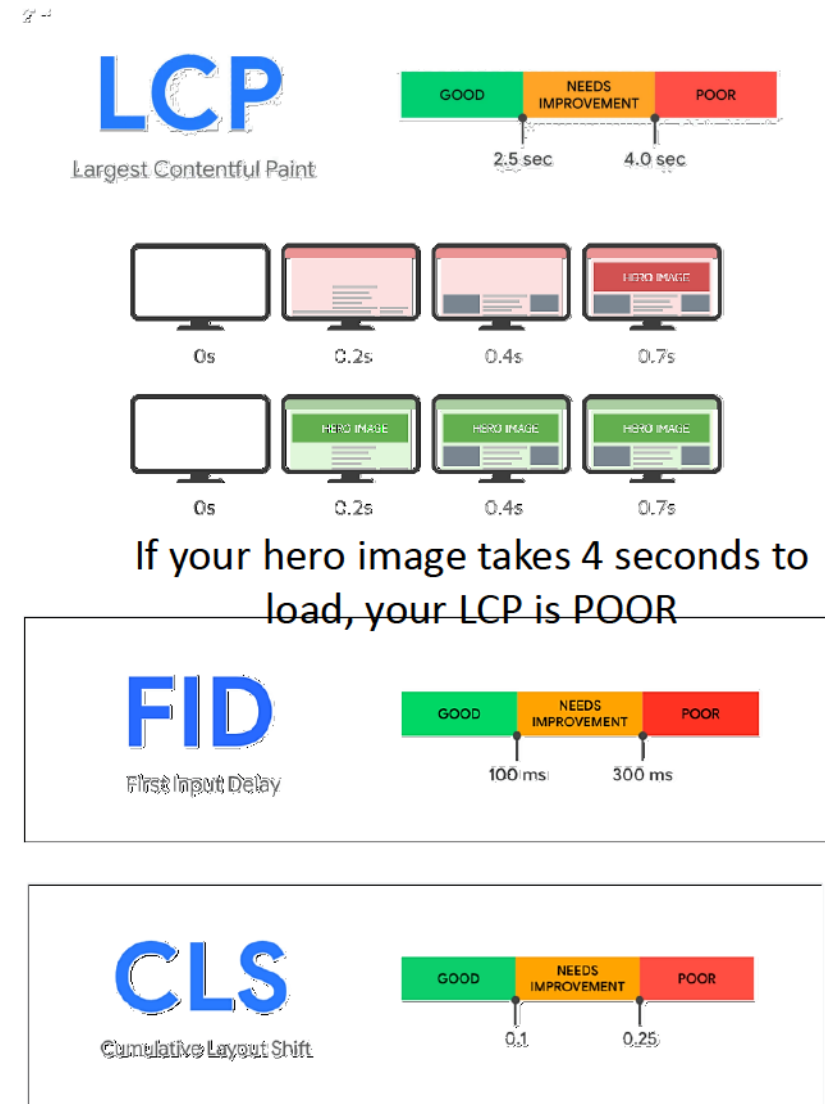
Modern Tools to Measure Performance:

- Google Lighthouse (built into Chrome DevTools)
- PageSpeed Insights (web.dev/measure)
- WebPageTest.org -- *Let's see exactly what Google measures... →*

Core Web Vitals - Google's Performance Metrics

What Google Measures:

- **LCP (Largest Contentful Paint):**
 - How fast does main content load
 - Good - nothing to do here = LCP of 1.2 sec. or less.
 - OK, but consider improvement = LCP between 1.2 and 1.67 sec.
 - Longer than recommended = LCP between 1.67 and 2.4 sec.
 - Much longer than recommended = LCP higher than 2.4 sec.
 - **Good: <2.5 seconds**
- **FID (First Input Delay) → Interaction to Next Paint (INP)**
 - How quickly can users interact?
 - **Good: <100 milliseconds**
- **CLS (Cumulative Layout Shift)**
 - Does content jump around while loading?
 - **Good: <0.1**
- **How to Check: Run Google Lighthouse on your site**



Chrome Dev Tools to Identify the LCP

The image shows a Chrome browser window with the CDC website open. The Chrome DevTools Performance tab is active, showing a recording of the page load. A red arrow points to the 'Performance' tab in the DevTools toolbar. A red box highlights the instructions for recording the page load.

Performance Tab

Click the record button  or hit **⌘ E** to start a new recording.

Click the reload button  or hit **⌘ ⇧ E** to record the page load.

After recording, select an area of interest in the overview by dragging. Then, zoom and pan the timeline with the mousewheel or **WASD** keys. [Learn more](#)

Chrome Dev Tools to Identify the LCP

The image shows a Chrome browser window displaying the CDC website. The Chrome DevTools Performance tab is open, showing a timeline of the page load. A red box highlights the 'Performance' tab in the top bar. Another red box highlights the 'Record' button (a circular arrow icon) in the top left of the Performance panel. The timeline shows various events: Network (w...), Frames (18.3 ms), Timings (FP, FCP, DCL, LCP), and Experience (Layout Shift, La...ft). A red arrow points from the 'LCP' marker in the Timings section to the image element in the browser window, which is labeled 'img.card-img-top.img-fluid.p-0 343 x 190.55'. The image shows an elderly couple smiling.

Don't worry about all the colors yet!
Focus on the LCP marker (labeled in green)

Measuring Website Success (Beyond Speed)

- **For E-Commerce Sites:**
 - **Conversion Rate:** % of visitors who buy
 - **Average Order Value:** Revenue per transaction
 - **Cart Abandonment Rate:** % who add items but don't checkout
- **For Content Sites:**
 - **Bounce Rate:** % of single-page visits
 - **Time on Page:** How long users engage
 - **Pages per Session:** Depth of exploration
- **For All Sites:**
 - **Mobile vs Desktop Traffic:** Who's your audience?
 - **Traffic Sources:** Where do visitors come from?
- **Tool:** Google Analytics 4 (GA4) tracks all of these

Factors Impacting Your Website Load Time

What You Can Control/Developers Can Optimize

1. Caching (Store data locally).
2. Image optimization (Use WebP, compress files).
3. Code efficiency (Minimize CSS/JS).
4. HTTP requests (Combine files).
5. Core Web Vitals optimization: loading speed, interactivity, and visual stability.
6. Third-party scripts (Load async).

What You Can't Control (But Should Know)

1. Hosting and server performance.
2. Network quality.
3. User's device.
4. Mobile optimization: Design for the worst-case scenario (slow phones, 3G networks)

Hands-On Exercise: Test Your Favorite Website

Instructions:

1. Open Chrome DevTools (F12)
2. Go to "Lighthouse" tab
3. Run audit on any 3 websites (e.g., amazon.com, CNN/BBC, Galala university website) or (Galala, Arizona and Ain Shams universities)

What to Look For:

- Performance Score (0-100)- LCP, FID, CLS values
- Opportunities list (what's slowing the site down?)

Discussion Questions:

- Which site performed best? Why?
- What were the biggest performance issues?
- How would you fix them?

Screenshot your Lighthouse report – submit it during the next lab!

Web Site Organization

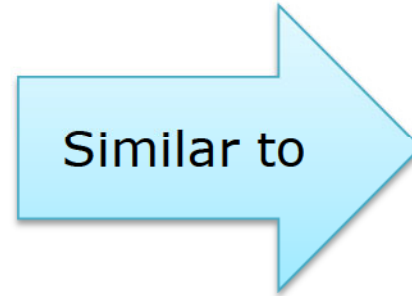
How is your site STRUCTURED?

Web Site Organization

- Hierarchical

- Linear

- Random
(sometimes called Web Organization)



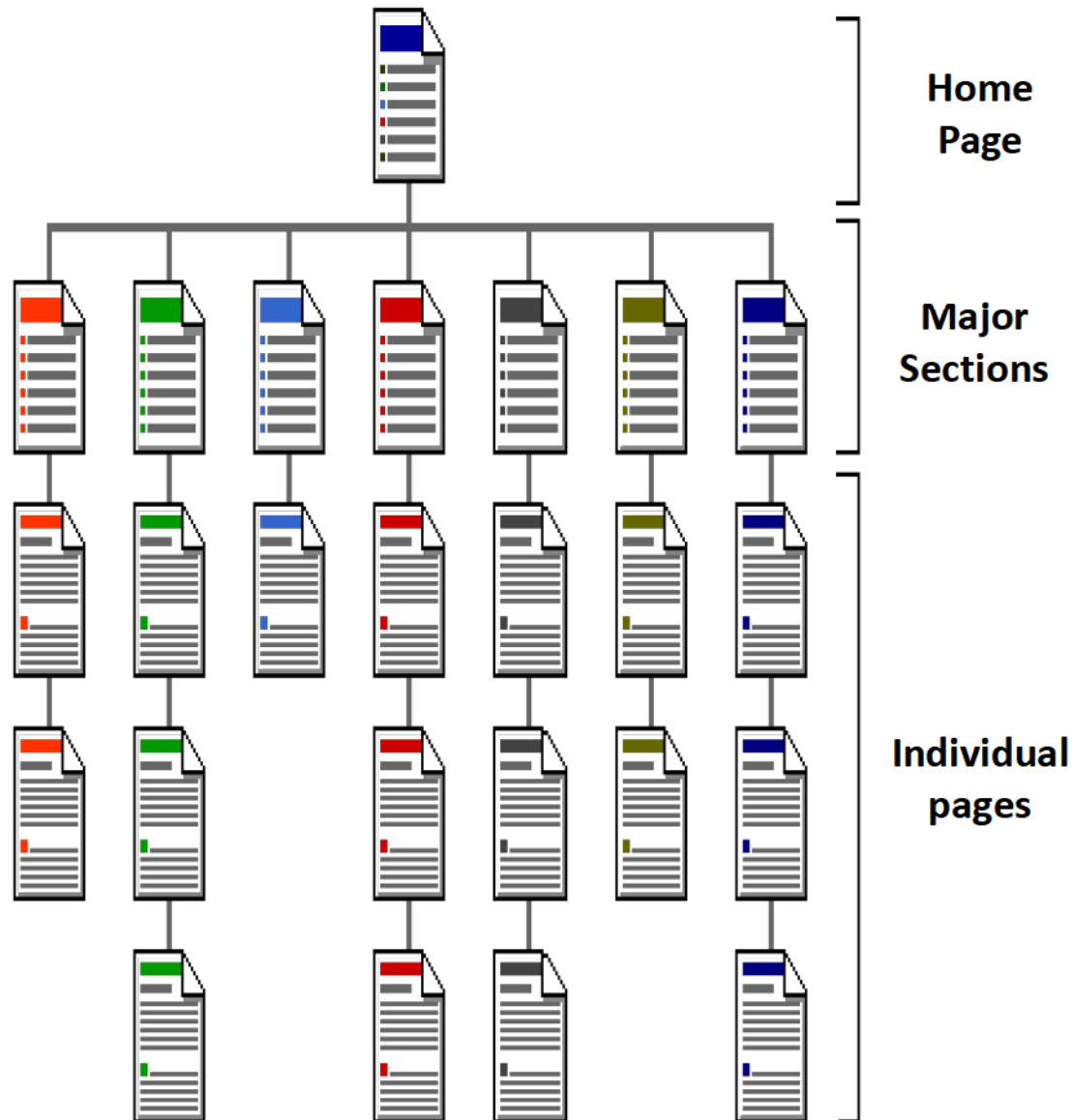
- File-system

- Tutorial wizard

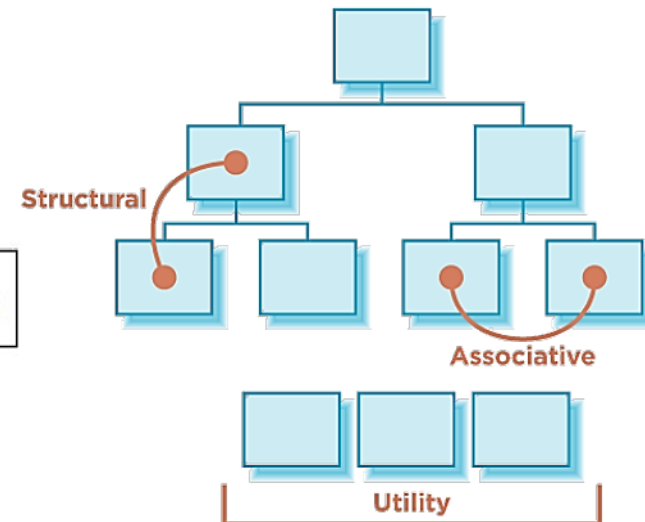
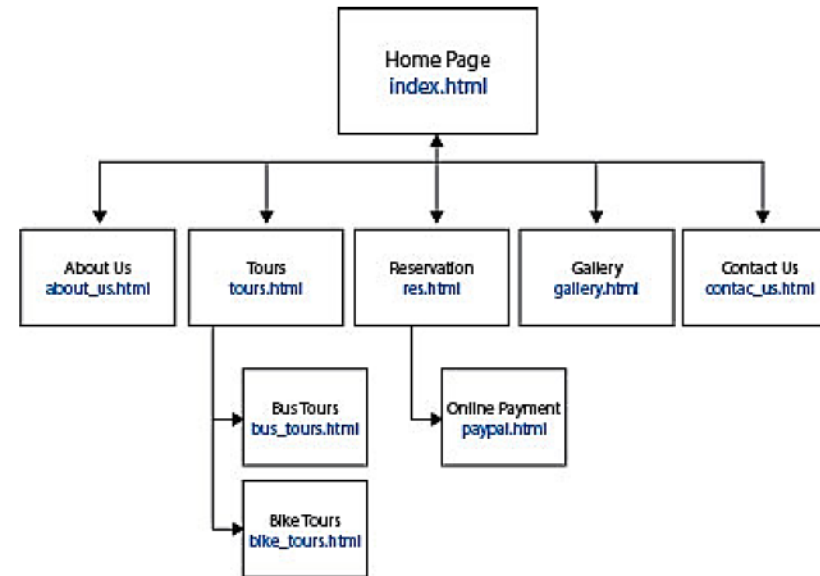
- Portfolio

90% of websites use Hierarchical organization

Hierarchical Organization

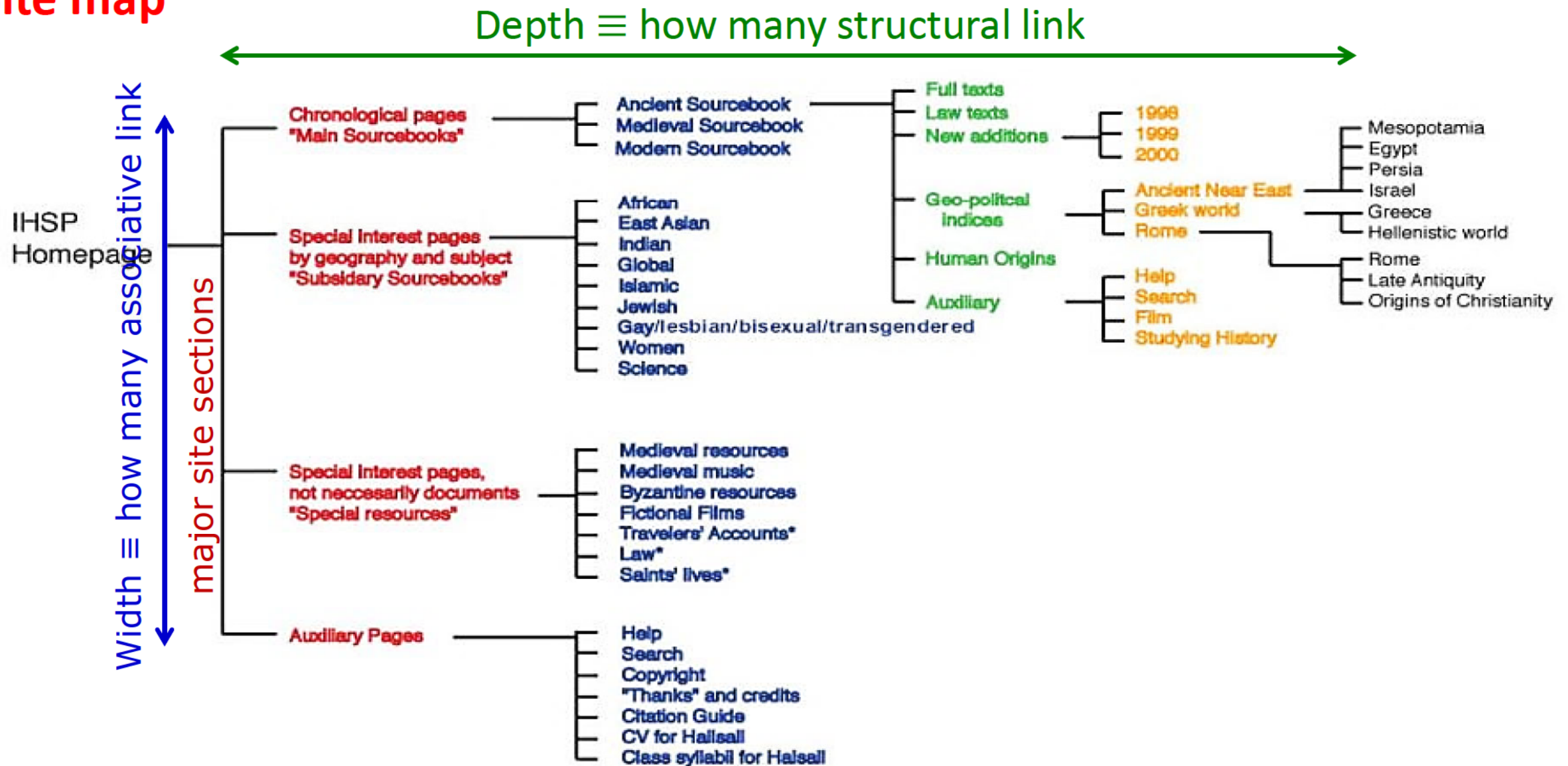


- ❑ Often used for commercial and corporate Web sites
- ❑ Start by clearly defined **home page**
- ❑ In home page **add Navigation links** to **major site sections**

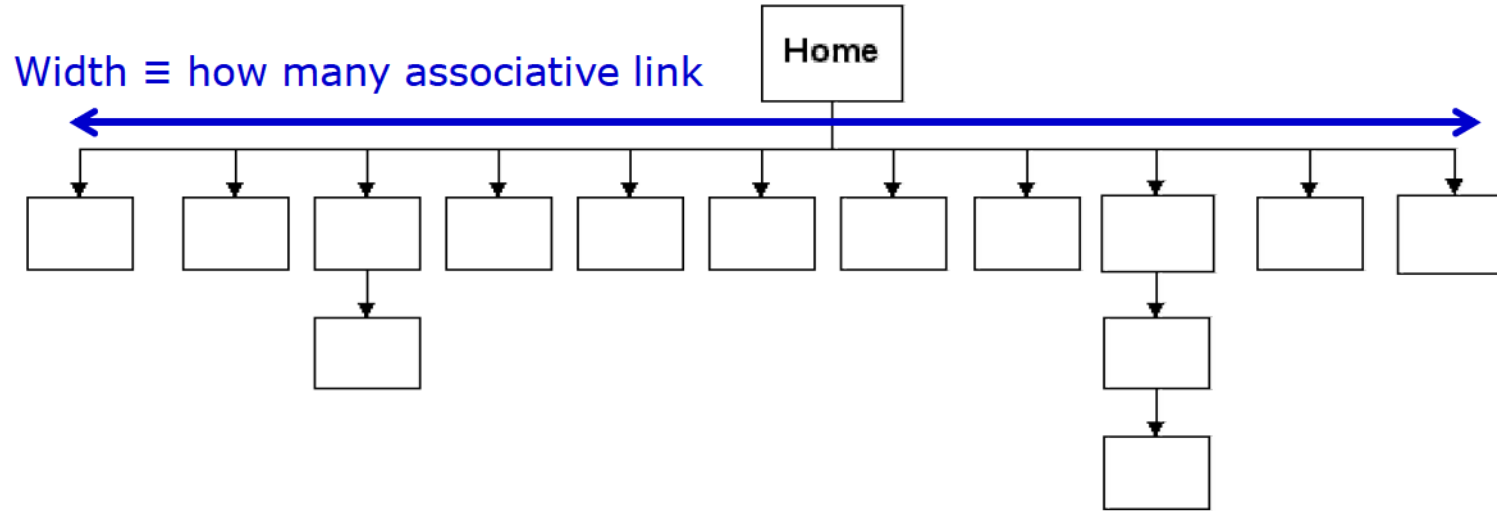


Hierarchical Organization: Site Map

Site map

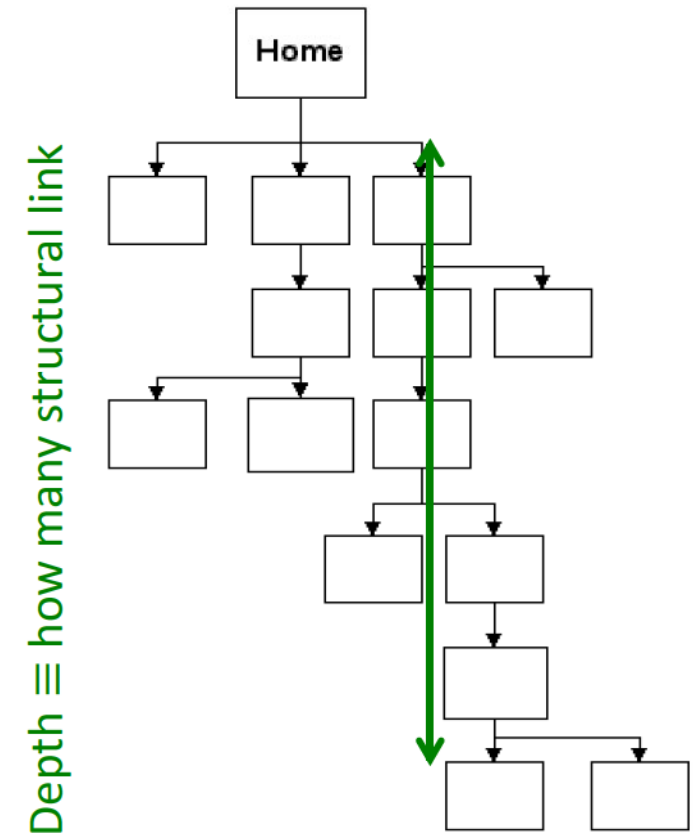


Hierarchical Organization Design Rules



Organization is **not too shallow**.

- # major sections: **7 ± 2**
 - ▣ Based on Miller's Law: humans can hold 7 ± 2 items in working memory.
- Don't use more than 9 navigation links on a page.
 - ▣ Reduces cognitive load and decision paralysis



- Organization is **not too deep**.
- User Interface "**Three Click Rule**"
 - **MYTH**: Users leave if they can't find content in 3 clicks
 - **REALITY**: Users stay if each click feels like progress
 - **BEST PRACTICE**: Make every click obvious and purposeful

Hierarchical example

yahoo!finance

Search for news, symbols or companies



News

Finance

Sports

More ▾



Mail

Sign in

My Portfolio

News

Markets

Research

Personal Finance

Videos

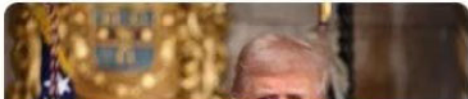
Streaming Now



Musk controversy is hitting Tesla at a bad time for EVs

Protests over Musk's role in the Trump administration are hurting the Tesla brand at a pivotal time.

Yahoo Finance • 13 hours ago **TSLA -0.49%**



Why Intel could be worth more than \$200B if broken up
Yahoo Finance • 17 hours ago
NVDA +0.40% **INTC +16.06%**

Musk's X in talks to raise money at a \$44 billion valuation
Reuters • 1 hour ago

Saylor's Strategy to raise another \$2B for bitcoin, cut workforce
Bloomberg • 10 hours ago

Oil holds onto gains with OPEC+ and Russian supply in focus
Bloomberg • 2 hours ago
CVX +0.53% **BZ=F +0.78%**

Why the White House claims Musk isn't really running DOGE
Yahoo Finance • 10 hours ago

Latest
Wix.com profit beats estimates, sees \$2 billion revenue in 2025
Reuters • 23 minutes ago **WIX +1.82%**

Indonesia Holds Key Rate Unchanged as Rupiah Back in Focus
Bloomberg • 24 minutes ago

BP Weighs Sale of Castrol Lubricants Unit Amid Elliott Push
Bloomberg • 27 minutes ago **BP +0.89%**

Glencore Reports Profit Slide as Key Earner Coal Slumps
Bloomberg • 30 minutes ago
GLNCY -0.79%

Europe Stock Rally Pauses on Tariffs, Geopolitics: Markets Wrap
Bloomberg • 35 minutes ago **^DJI +0.02%**

UK Inflation at 10-Month High as VAT Hits Private Schools
Bloomberg • 38 minutes ago

Gold Steadies Near Record as Mounting Trade Risks Loom Large

Quote Lookup



U.S. markets open in 5h 33m

US Europe Asia Rates Commodities

S&P Futures 6,152.50 +5.75 (+0.09%)	Dow Futures 44,671.00 +28.00 (+0.06%)	Nasdaq Fu... 22,262.50 +30.25 (+0.14%)
Russell 20... 2,297.10 -0.40 (-0.02%)	Crude Oil 72.46 +0.61 (+0.85%)	Gold 2,954.90 +5.90 (+0.20%)

< >

MY PORTFOLIOS

Sign in to access your portfolio

Sign in

TOP GAINERS

Hierarchical example

SHARE A
VIRTUAL BOTTLE

BUY YOUR
OWN

SHARE
GALLERY

FIND
YOUR NAME

SHARE A
COKE® TOUR

WHAT'S IN
A NAME

MORE WAYS
TO SHARE

\$5

8 fl oz.
Bottle

Free Shipping!

For orders of 6
bottles or more.



Buy Your Own
Personalized Bottle

Buy Now



Celebrate Birthdays



Women's Soccer



Wedding Party Fun

Hierarchical example

US | English

Free shipping on small orders over \$50 for IKEA Family members only

33132 Miami, FL



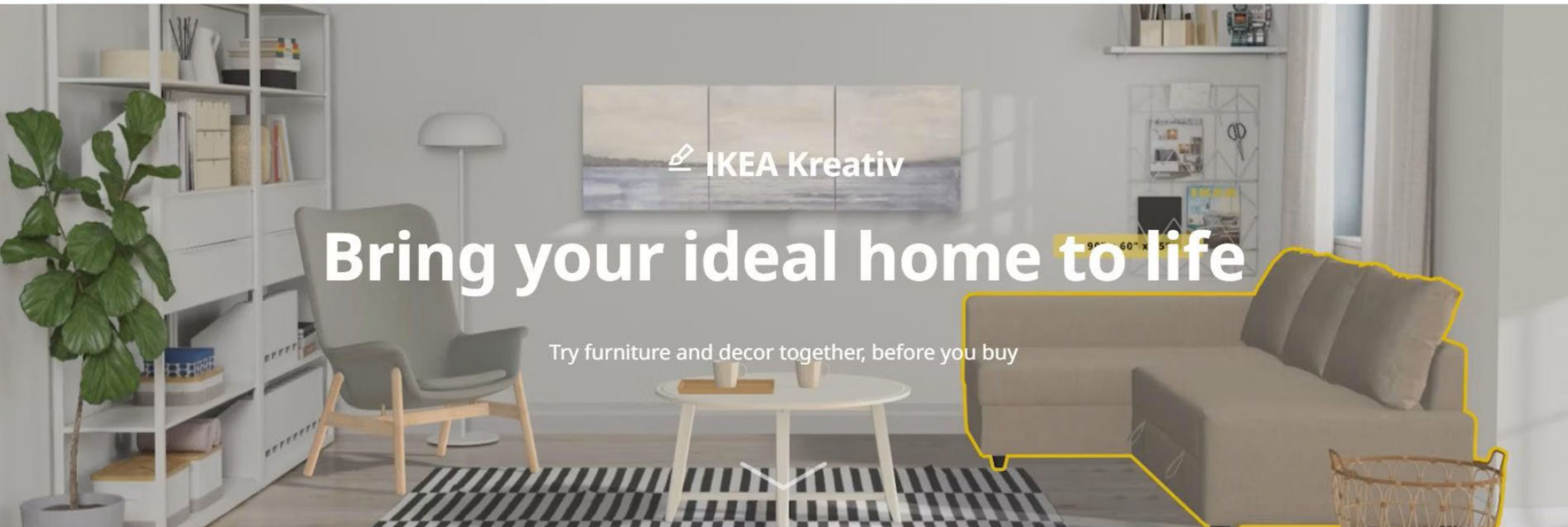
What are you looking for?



Hej! Log in or sign up



Products Rooms Deals Spring home ideas Home accessories Ideas & inspiration Design & planning Services & support IKEA for Business



Hands-On Exercise: Quick Challenge

Open Galala university's website. Identify:

1. How many main navigation sections? (Width)
2. What's the deepest page you can find? (Depth)
3. Does it follow the 7 ± 2 rule?

Discuss with a partner:

Would you restructure anything?

When NOT to Use Hierarchical Organization

Hierarchical Works Best For:

1. E-commerce sites (categories → products)
2. Corporate websites (about → services → contact)
3. Content-heavy sites (news, blogs)

Consider Other Patterns When:

- Content has no natural hierarchy (art portfolios)
- User journey is sequential (checkout process → Linear)
- Exploration is the goal (creative portfolios → Random)
 - Example: Behance.net uses Portfolio organization for designer profiles

Linear (Sequential) Organization

Linear (Sequential) Navigation: best for

- Tutorials and courses (Lesson 1 → Lesson 2 → Lesson 3, ...)
 - Checkout processes (Cart → Shipping → Payment)
 - Onboarding flows (Welcome → Setup → Start)
- Example: Duolingo lesson structure

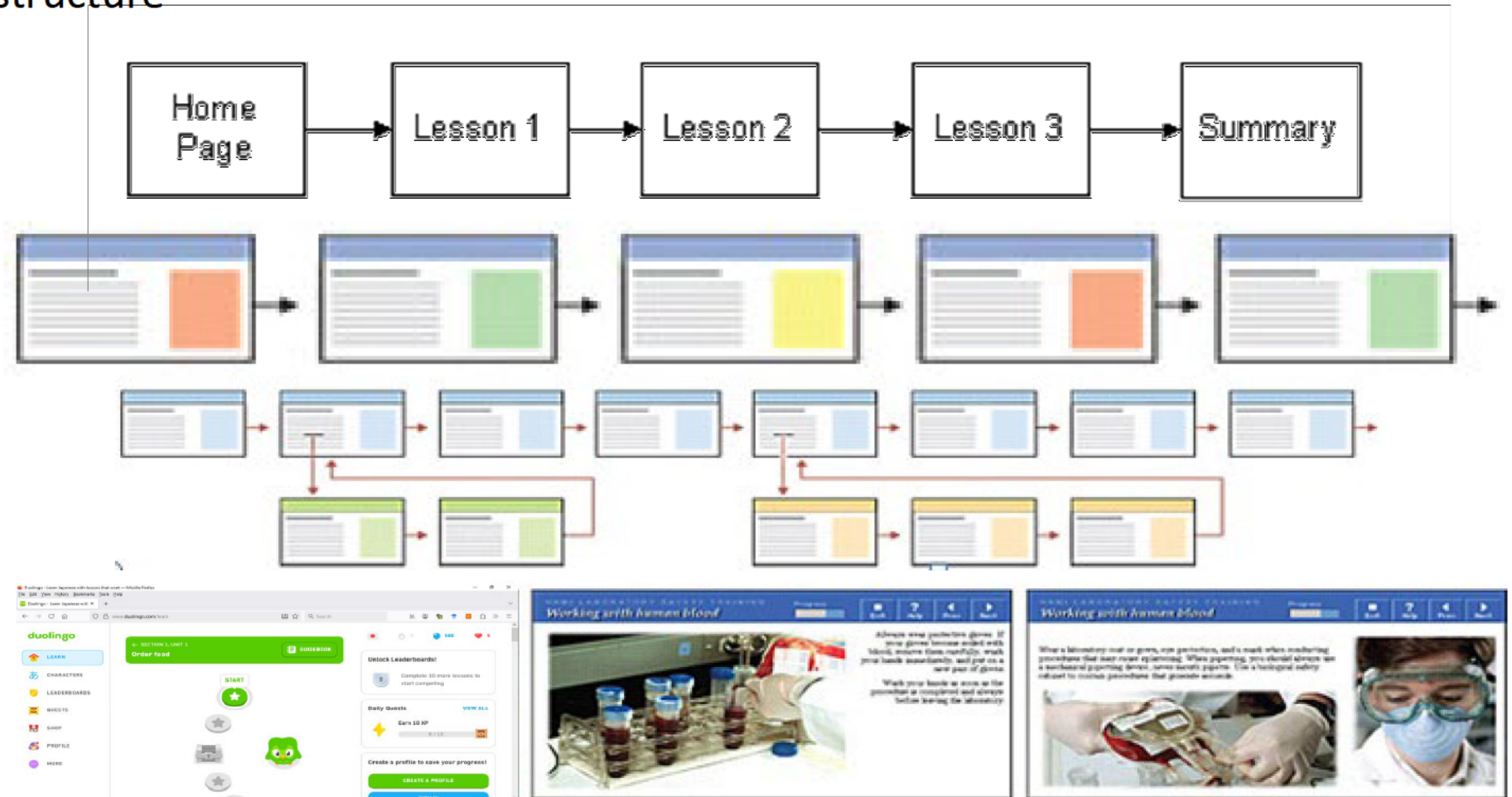
Warning:

Users can't jump ahead -
frustrating if misused!

Best Practice:

Show progress indicator
(e.g., "Lesson 2 of 5")

- Users tolerate linear flows better when they know how long they'll take



Random (Portfolio) Organization

Sometimes called “**Web**” Organization

Usually there is **no clear path** through the site

Still provide a "Home" escape route!

Best For:

- Artist portfolios (free exploration)
- Creative agencies (discovery-focused)
- Museum websites (non-linear browsing)

Example 1: awwwards.com (web design showcase)

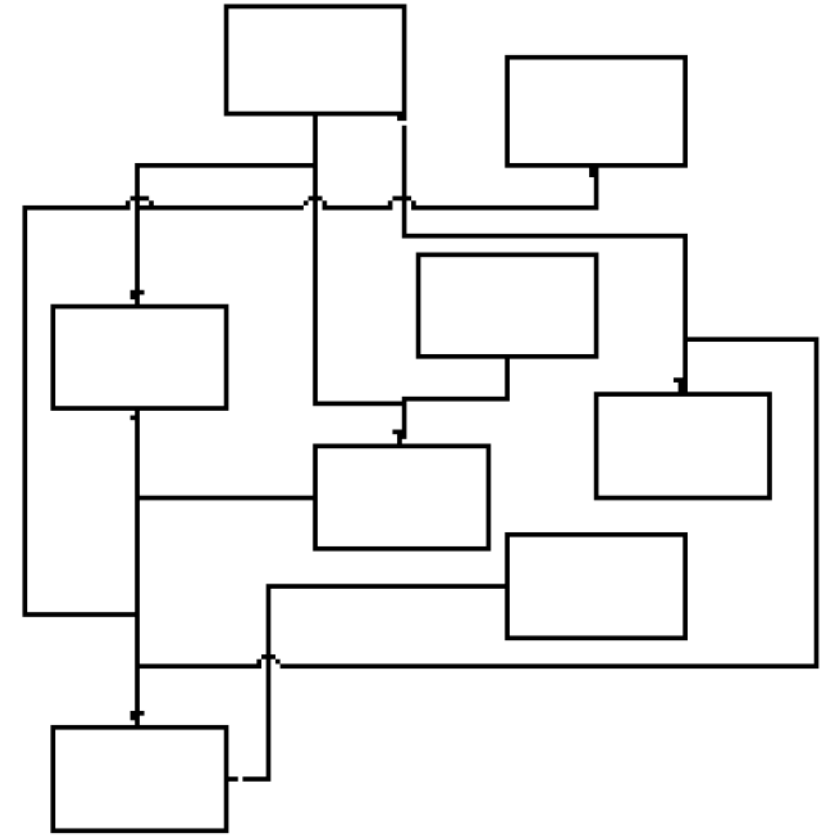
Example 2: Behance.net (designer portfolios)

- Each project can be viewed in any order
- Related projects links create web-like connections

Warning: Only use when exploration IS the goal

Don't use for: E-commerce, news sites, documentation

Do use for: Portfolios, art galleries, experimental sites



Hands-On Exercise: Test Your Favorite Website

Instructions:

1. Open these three websites:
 - ▣ [amazon.com](https://www.amazon.com)
 - ▣ [duolingo.com](https://www.duolingo.com)
 - ▣ [behance.net](https://www.behance.net)
2. With a partner, identify
 - ▣ Primary organization pattern
 - ▣ Any secondary patterns used
 - ▣ Why that pattern fits the site's purpose
3. Share: One insight per pair (whole class discussion during the Lab)

Website Navigation

**How do users MOVE through your structure?
What makes navigation EFFECTIVE?**

Types of Navigation: How do users MOVE through your structure?

❑ Graphics-based

- Good: Icons with text labels
- Avoid: Graphics-only navigation (accessibility issues)

❑ Text-based

❑ Interactive Navigation Technologies

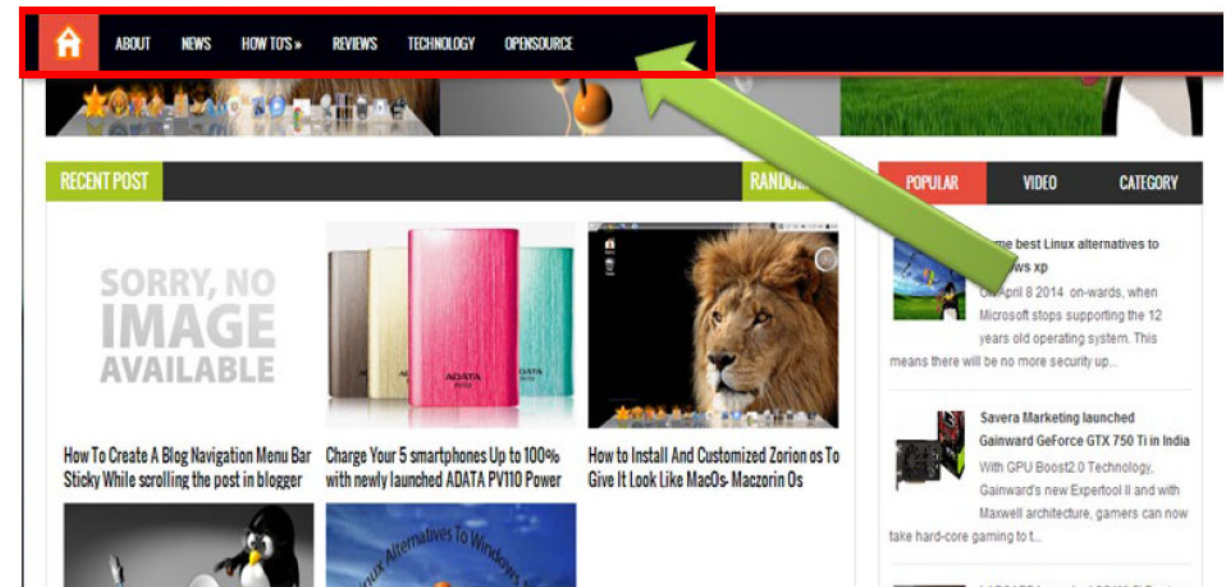
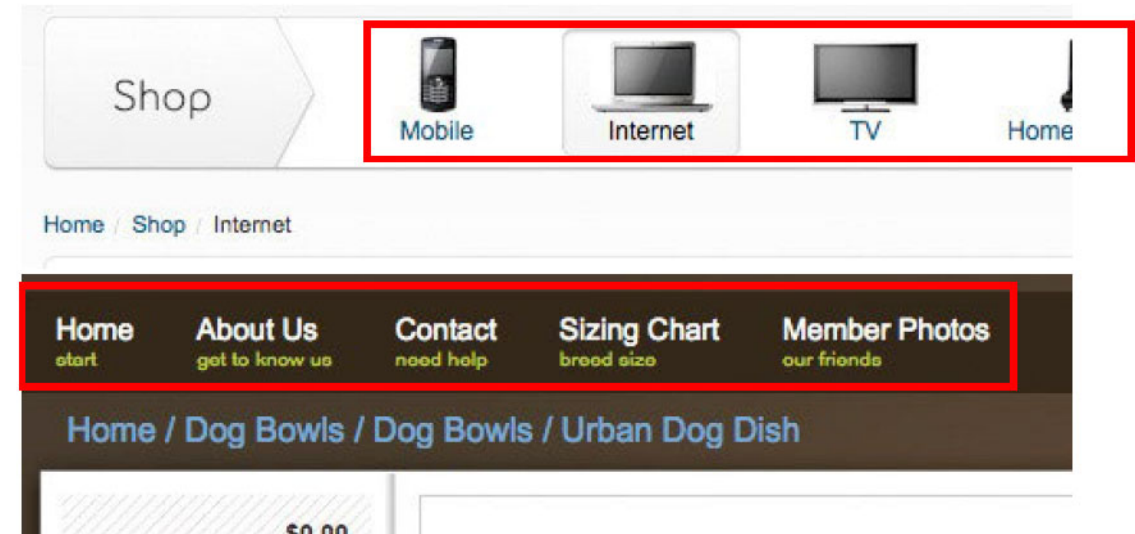
- CSS hover effects
- JavaScript dropdown menus
- React/Vue component-based navigation

❑ Modern Standard:

- Hybrid Approach: Icons + Text = Best of both worlds
- Example: GitHub
 - icon + label on desktop, icon-only on mobile

Accessibility Tip

- ALWAYS provide text alternative for visual navigation
- Use semantic HTML (<nav>, ,)
- Test with keyboard-only navigation
- Ensure sufficient color contrast (WCAG AA: 4.5:1)



Web Site Navigation Best Practices

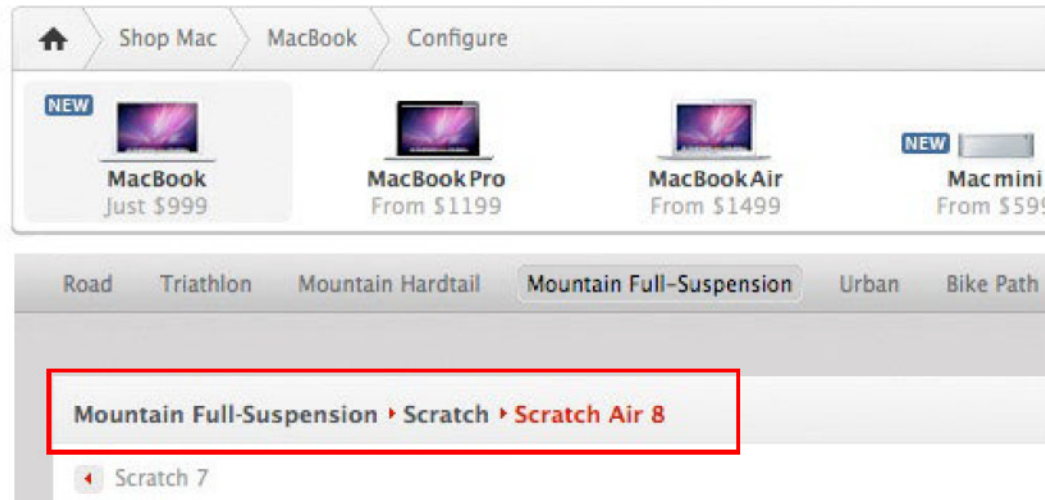
What makes navigation **EFFECTIVE**

- Provide clearly labeled **navigation** in the **same location on each page**
- Most common – **across top or down left side**
- Provide “**breadcrumb**” navigation: **Breadcrumb** navigation shows the **user's path** to their current location.



Accessibility First:

- Keyboard navigable (Tab key)
- Screen reader friendly
- Clear focus indicators



Web Site Navigation Best Practices

- Use a **Table of Contents** (with links to other parts of the page) for large # pages.
 - Example: Wikipedia articles (jump to sections) , Sticky TOC on sidebar = even better UX
- Consider **breaking long pages** into multiple shorter pages using **Linear Organization**.
- Large sites may benefit from a **site map** or **site search** feature

Hands-On Exercise: Audit Galala website navigation

This is how professional UX designers evaluate navigation!

Check for:

1. Consistent placement across pages?
2. Breadcrumbs present?
3. Max 7 ± 2 main navigation items?
4. Keyboard accessible?
5. Mobile-responsive?

Class discussion during the Lab

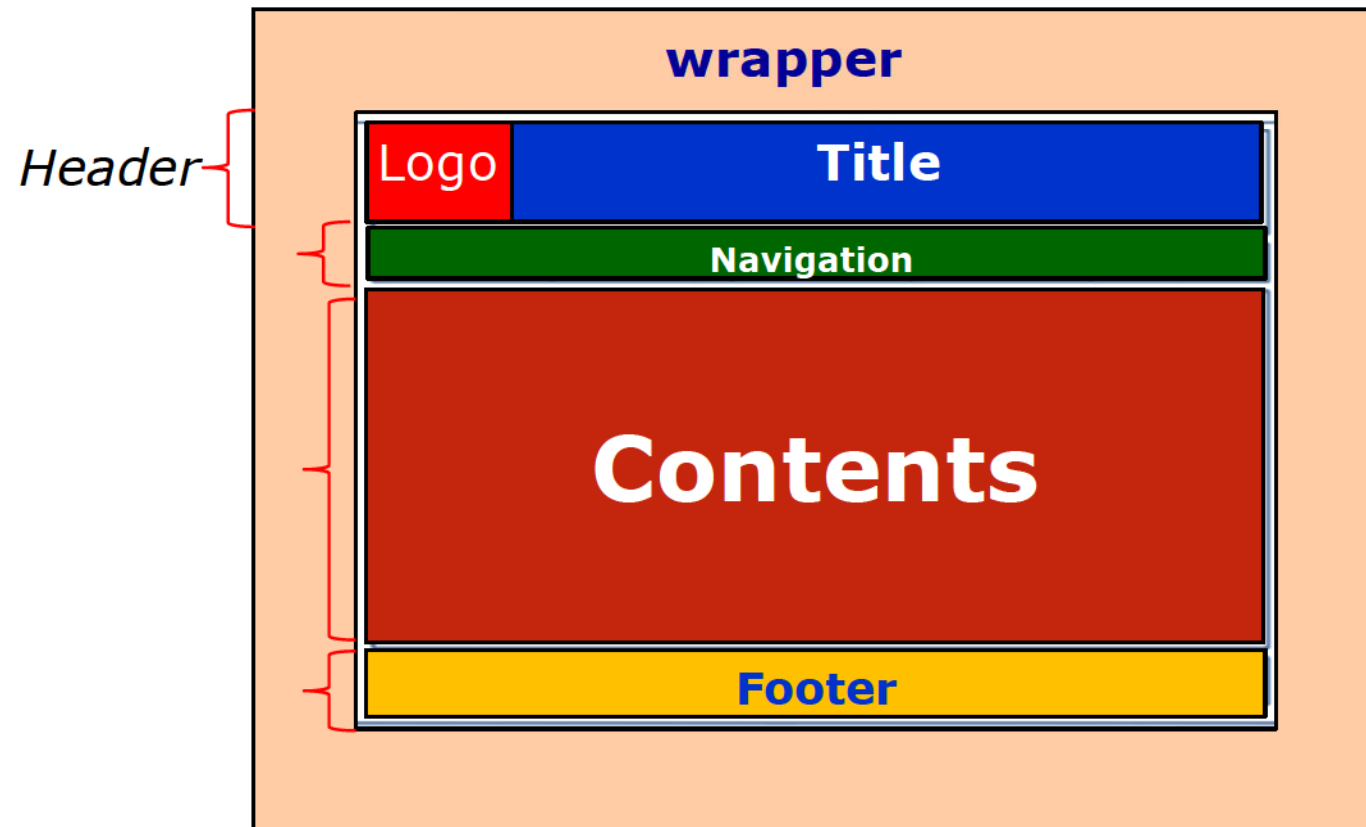
- Rate: 1-5 stars
- Share what would you improve?

Web Pages Layout Design

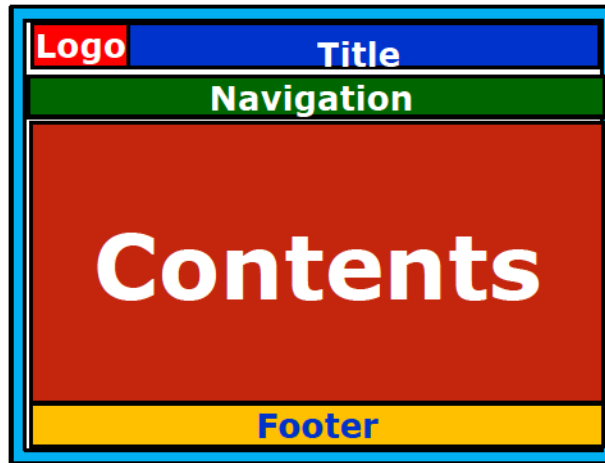
Web Page Design Page Layout: Wireframe

- A sketch of **blueprint** of a **Web page**
- Shows the **structure** of the **basic page elements**, including:

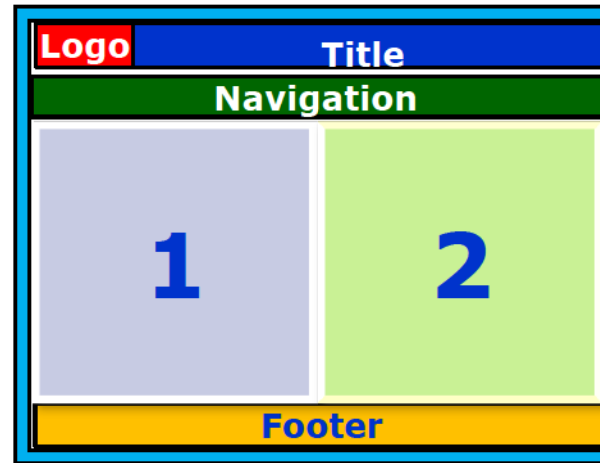
- Logo
- Title
- Navigation
- Content
- Footer



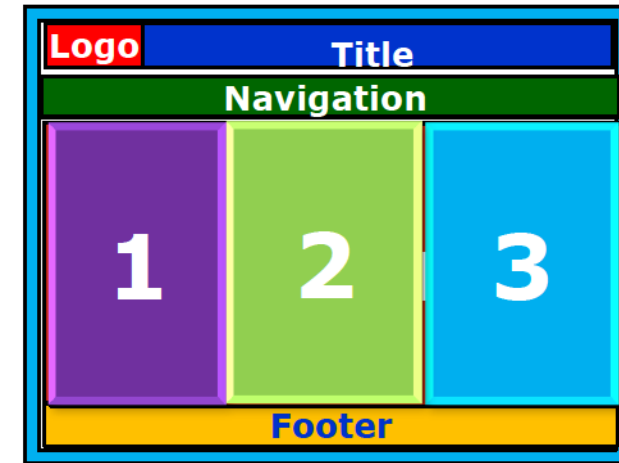
Page layout design: # columns



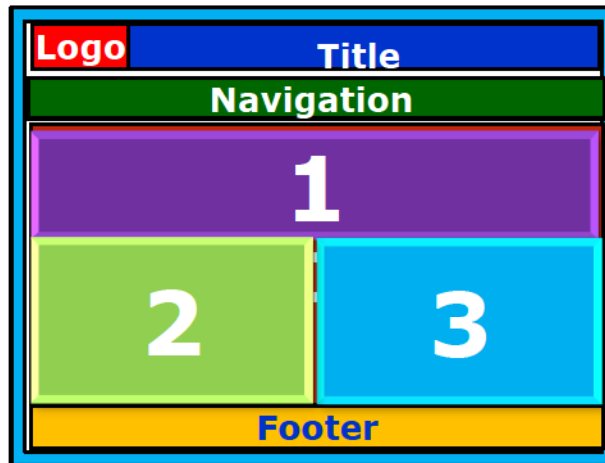
Single Column



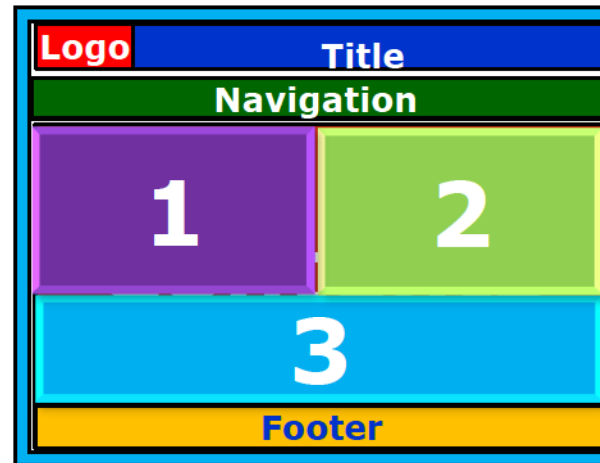
Two Columns



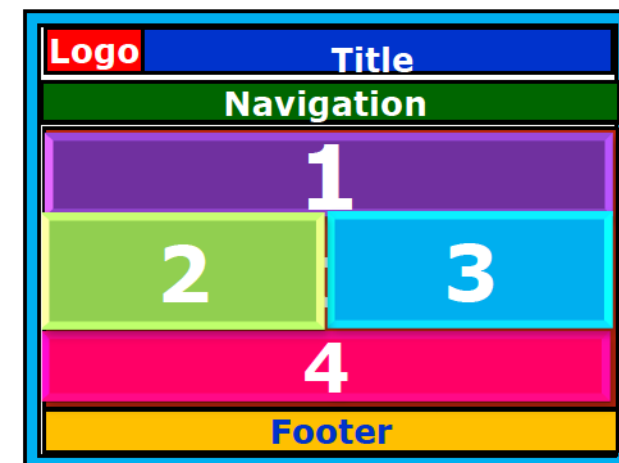
Three Columns



Three Columns



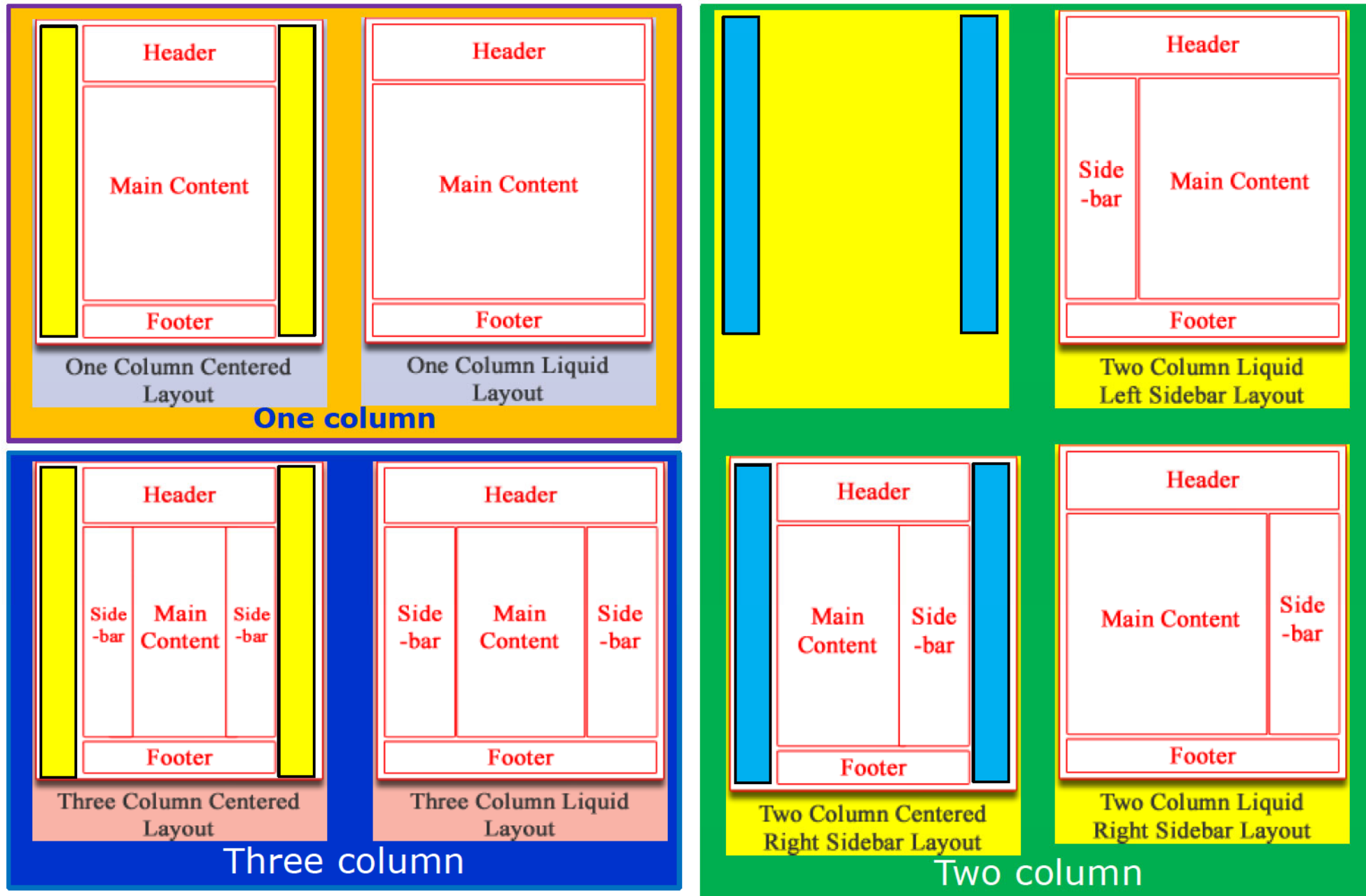
Three Columns



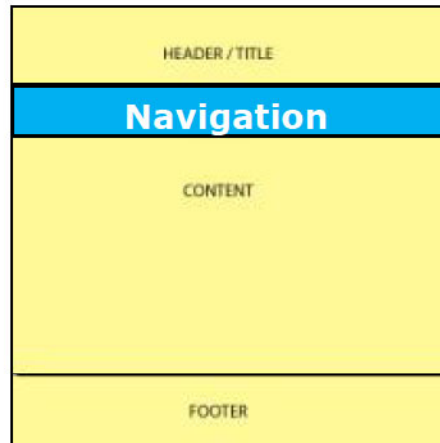
Four Columns

- People tend to **scan** web pages and decide whether they want to read them in full or not.
- The human eye scans a web page in a **Z shaped pattern**

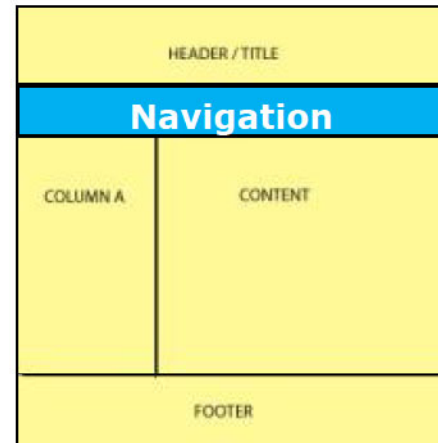
Webpage Layouts: centered vs liquid



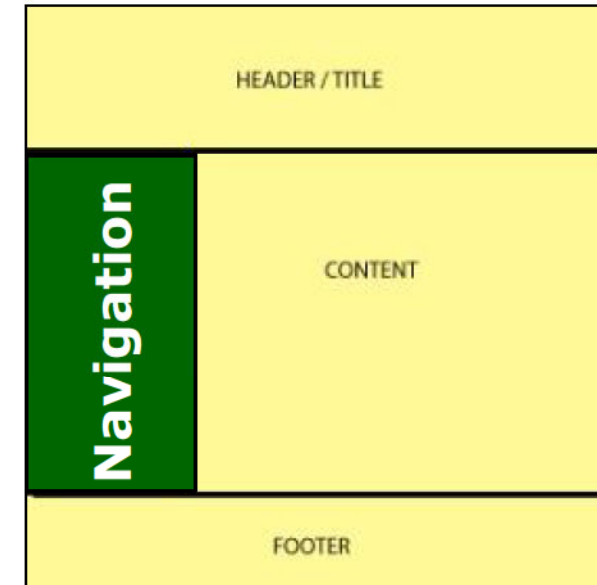
Webpage Layouts: Location of navigation bar



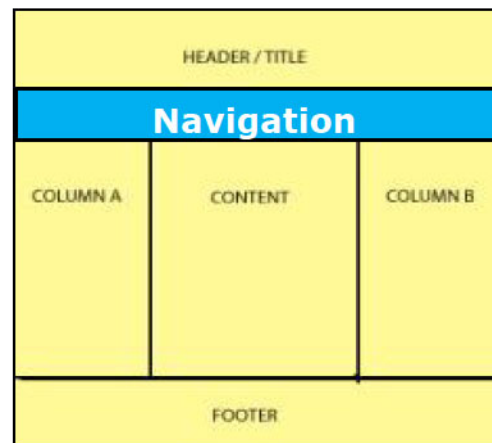
One Column With
Horizontal Menu Bar



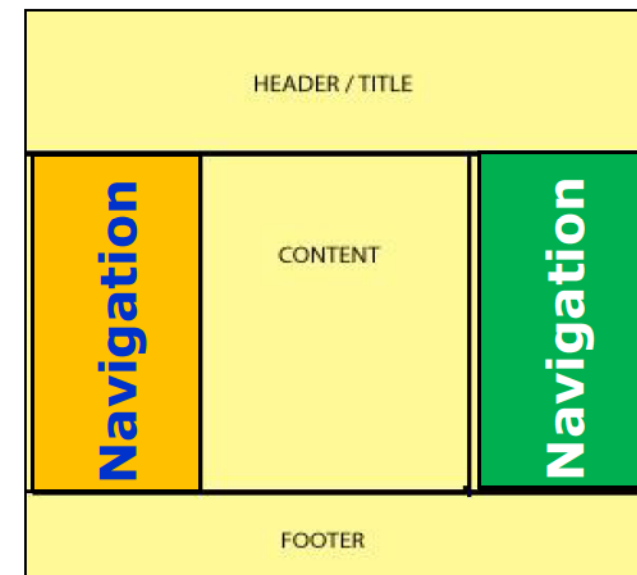
Two Column With
Horizontal Menu Bar



Two Column With **Vertical Menu**



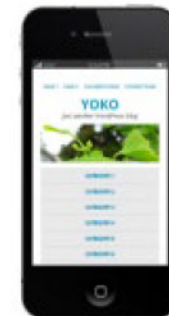
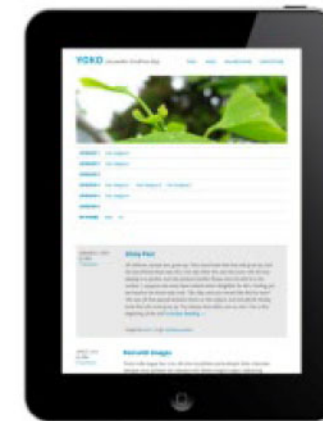
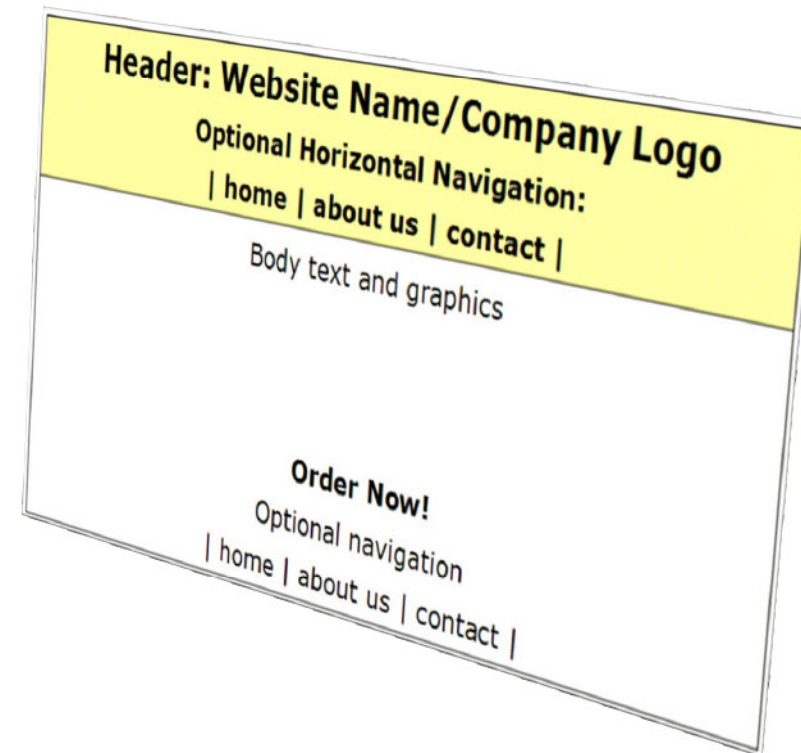
Three Column With
Horizontal Menu Bar



Three Column With **Vertical Menu**

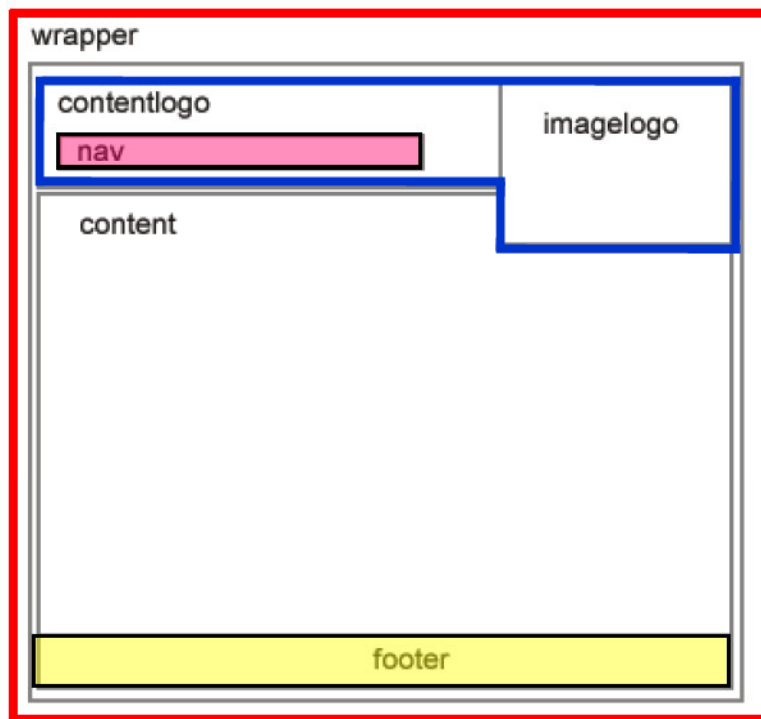
One Column Web Page

- With or without sidebars.
- This type of layout is commonly used to simply show the content, **no ads**, and **no extras**.
- The layout is good for **review sites** or
- **personal** or **corporate blogs**
- **Mobile applications**



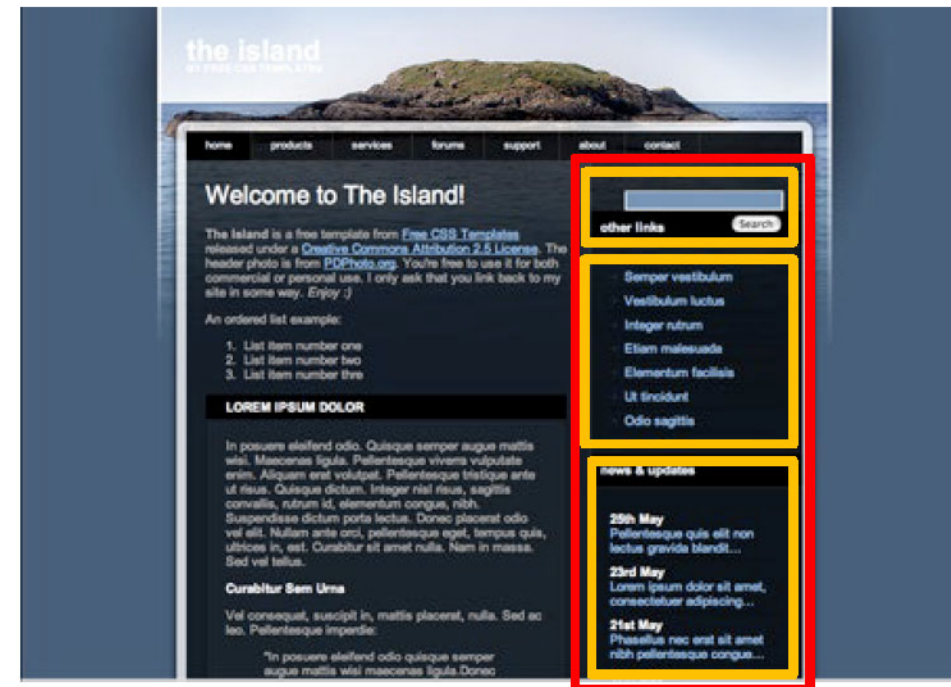
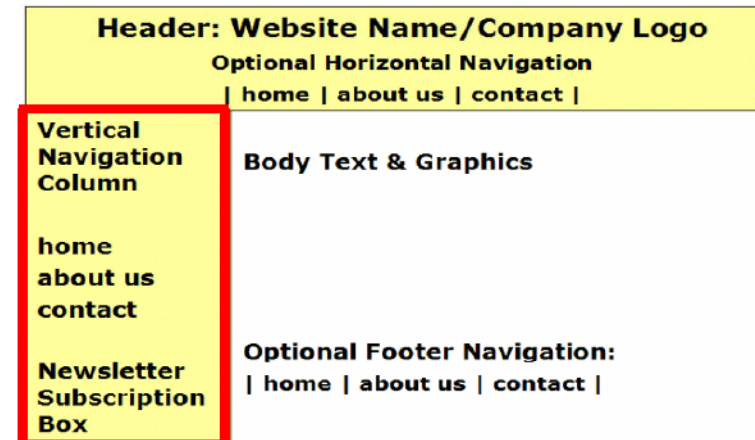
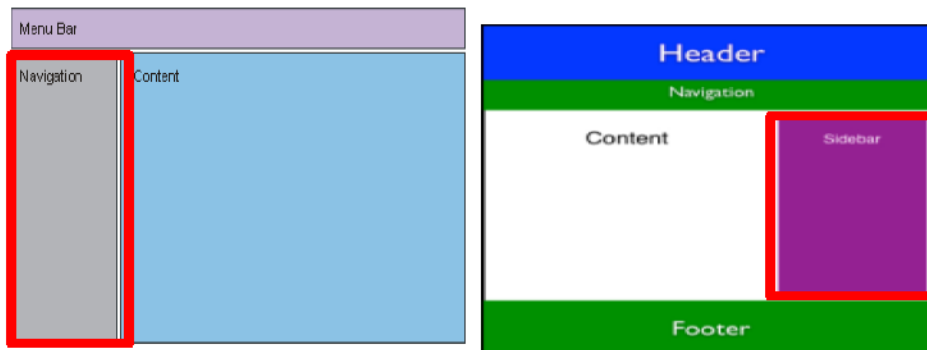
Single column Page Layout

Except for image logo,
all elements on this page follow normal (sequence) flow

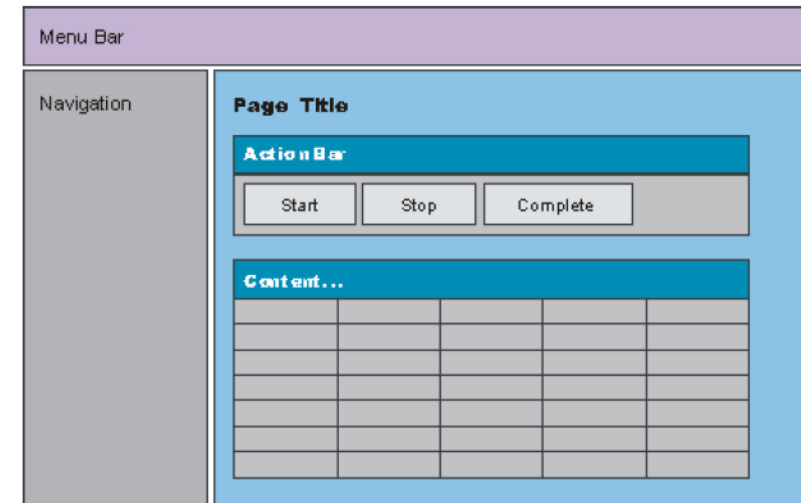
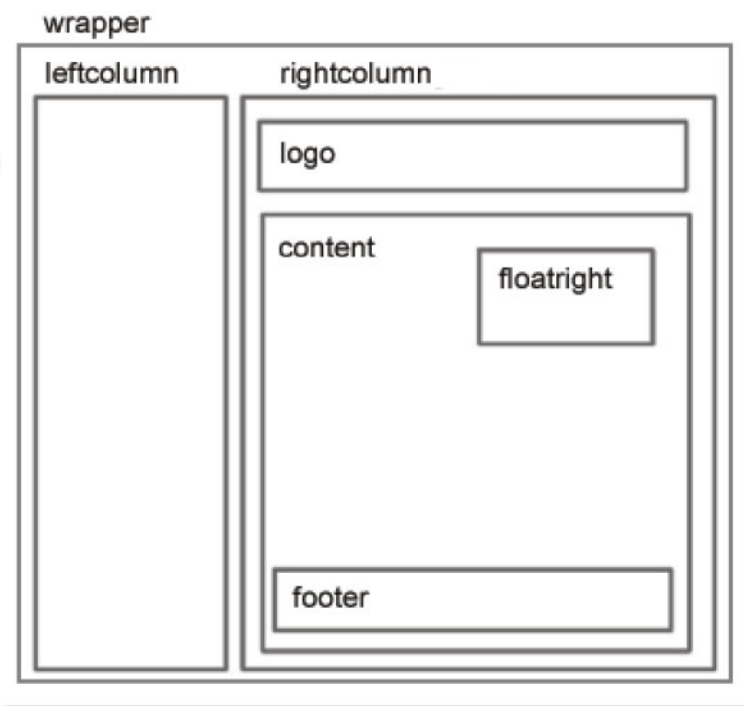


Two Column Web Page

- ❑ Generally for **E-commerce applications**
- ❑ This layout uses **one sidebar** that is either placed on
 - the **left side** or
 - on the **right side**.
- ❑ The sidebar contains:
 - different widgets used for
 - ❑ search bar,
 - ❑ easy navigation menus,
 - ❑ advertisements or
 - ❑ social bookmarking.

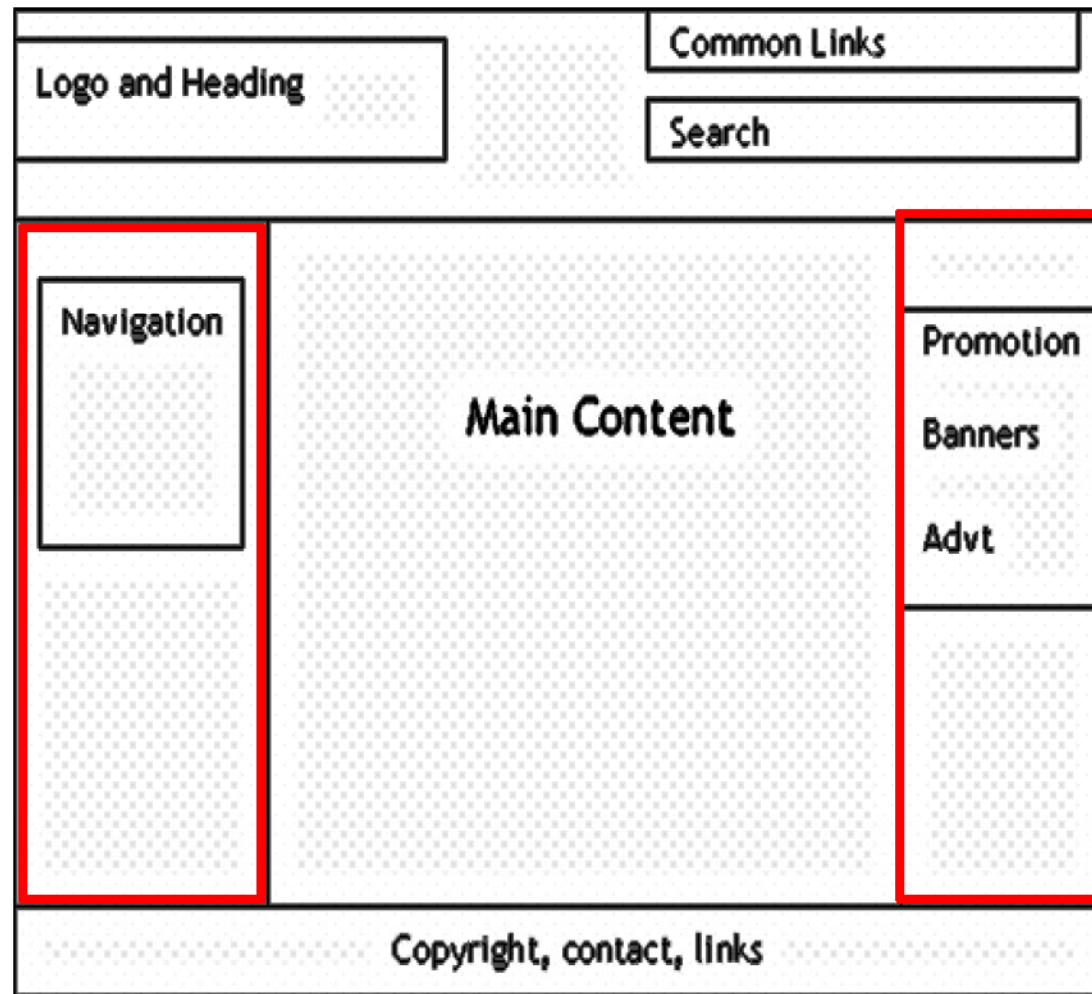


Two Column Page Layout

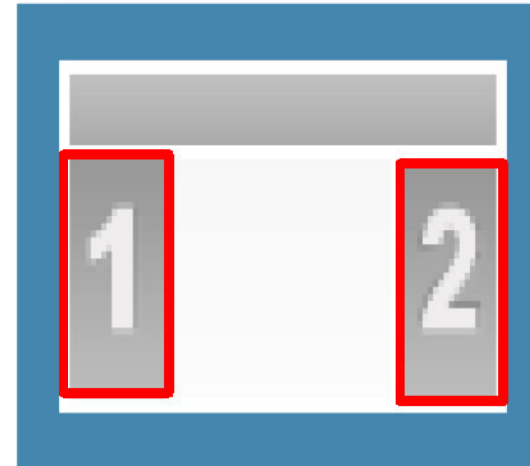


- ❑ wrapper contains the two columns – sets default background color
- ❑ Left-column navigation
- ❑ Right-column content
- ❑ floatright (flower photo)

Three Column Web Page



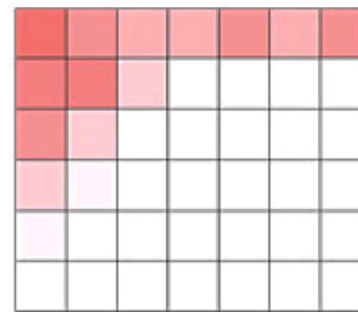
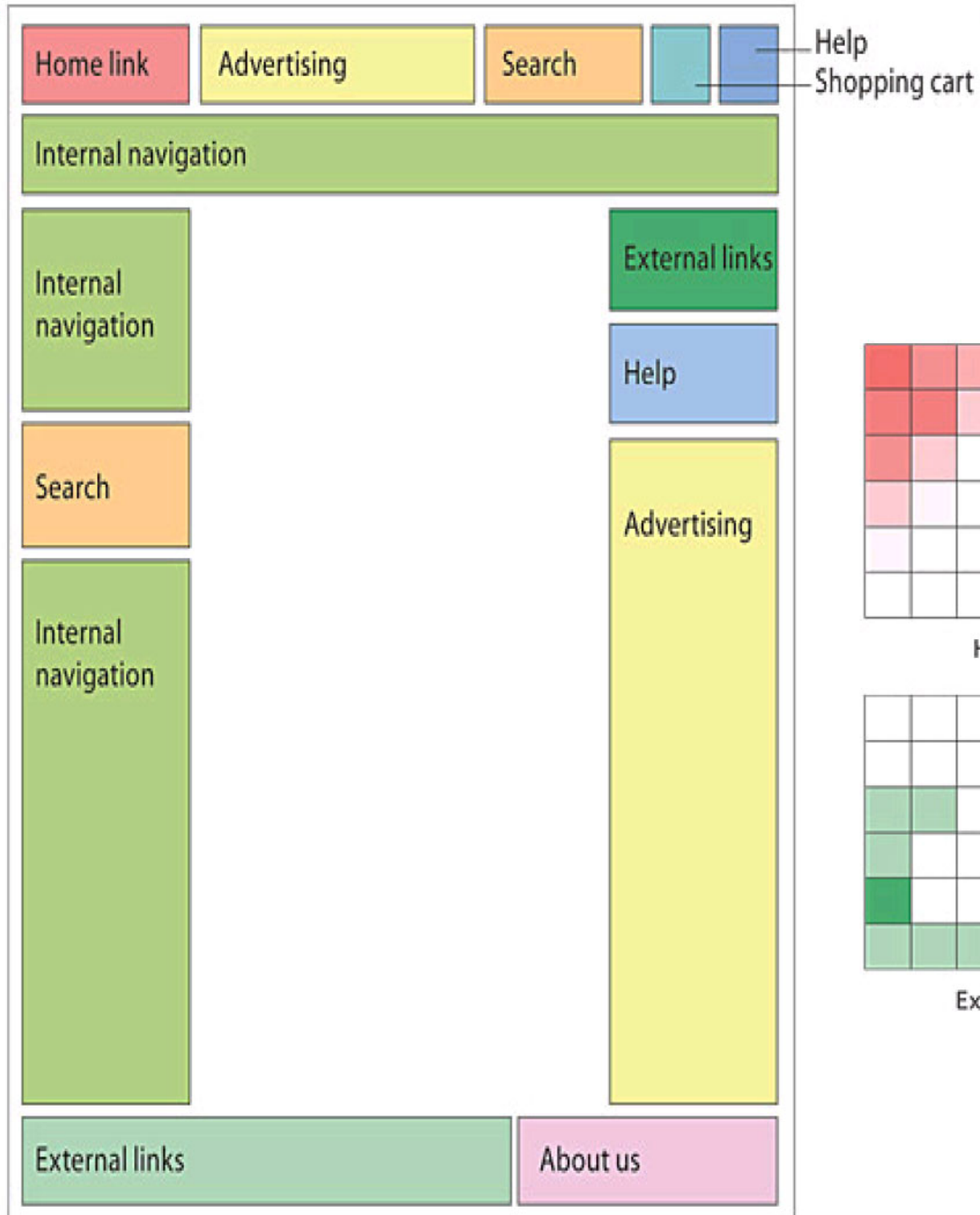
Three Column Web Page



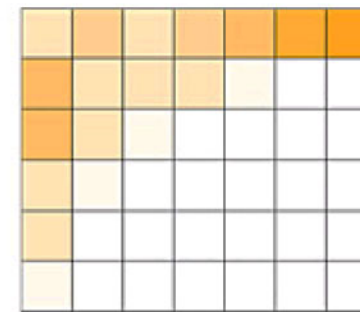
- This layout consists of **two sidebars**, usually one on the left side and one on the right side.
- sometimes **both sidebars** are on the right side.
- This is a typical layout **used for corporate websites**.



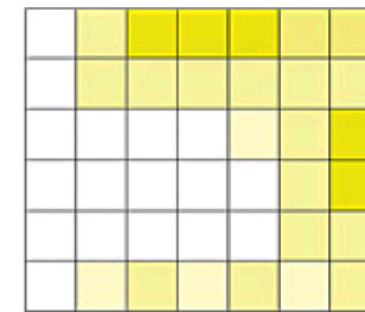
Wireframes



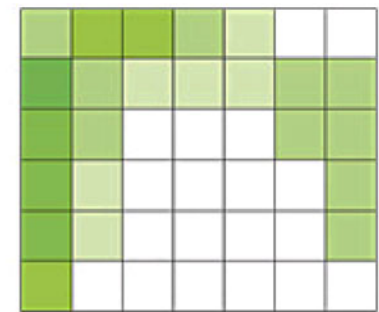
Home link



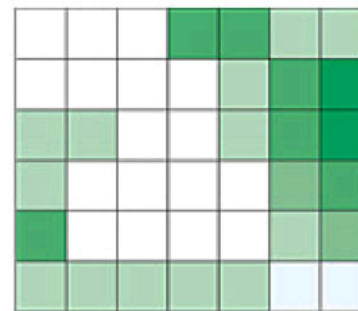
Search



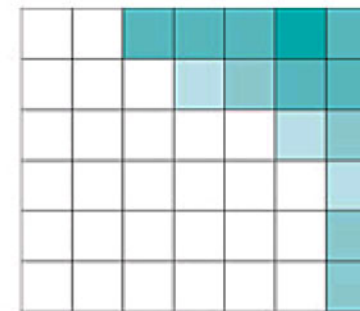
Banner ads



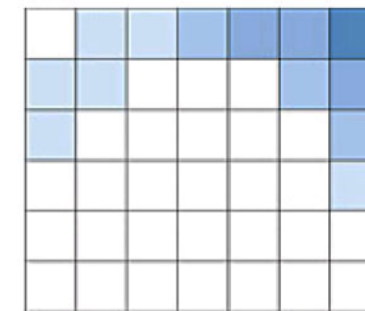
Navigation links



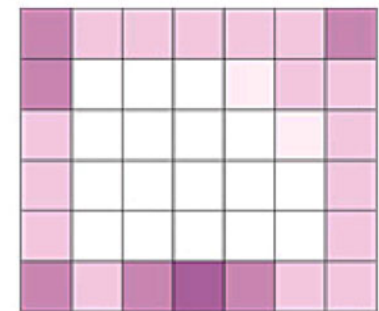
External links



Shopping cart

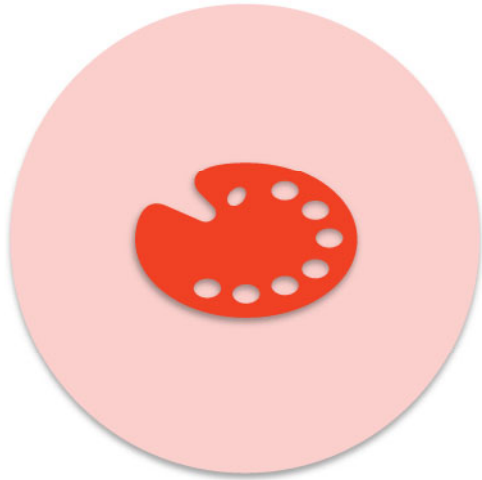


Help link



About us

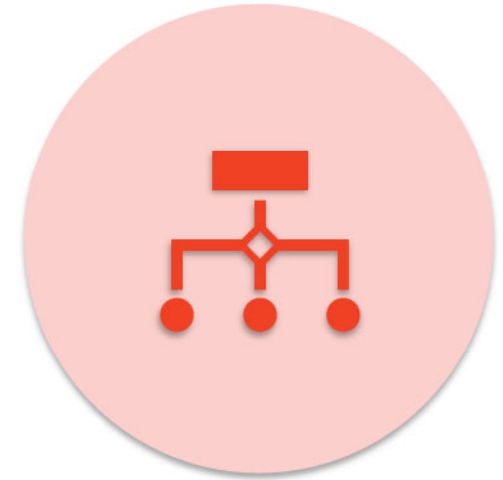
Add Hands-On Activities



Use paletton
to create a color palette, then
explain your choices



Sketch a wireframe
for an e-commerce homepage
for education homepage



Analyze 3 websites:
identify their layout type
(hierarchical/linear/random)

Questions?

